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“BECAUSE IT’S THERE!” English climber George Mallory reportedly said in 1924 when asked why he was trying to climb Mt Everest for the third time. By then, the farthest corners of the Earth had been explored, and only Everest was left. Mallory and fellow climber Andrew Irvine lost their lives in that bid.

A century later, that retort could sum up why the private sector has become a critical part of India’s space exploration plans, which grabbed eyeballs in 2014 with Mangalyaan. The Mars orbiter mission, its first interplanetary mission, cost less than Hollywood movies on science fiction, and the three Moon missions (2008, 2019 and 2023) have shown that the Indian Space Research Organisation (Isro) has come of age.

In 2020, the government allowed the private sector to reach for the stars, creating the Indian National Space Promotion and Authorisation Centre (IN-SPACE). Private companies swarmed into the sector.

Coming up: missions targeting other planets and celestial bodies, a Mars mission and lunar exploration. By 2030, India’s share of the global space economy will be four times what it was in 2021, going up to 8% by 2030 from 2% in 2021. By 2047, India’s share will be 15%.

Sam Hutchison, Co-founder

of Space Exploration and Research Agency (SERA), is pitching for democratised space travel. He says a few nations have dominated space exploration. Over 80% of all astronauts have come from three countries—the US, Russia, and China.

Hutchison says things are changing. SERA is leveraging innovative space technologies such as Blue Origin’s New Shepard rocket to build a community-driven platform that allows anyone, anywhere, to participate in space exploration and research. “As India’s national manned space programme advances, we are excited to include India in our upcoming mission,” Hutchison says. SERA will help India launch a citizen astronaut.

Private sector innovations, start-ups in satellite technology, space tourism and other commercial ventures are expected to fuel India’s space economy. Invest India, the national investment promotion and facilitation agency, says India’s space economy was worth around ₹6,972 crore (\$8.4 billion) in 2023 and accounted for 2-3% of the global space economy.

This will grow to around ₹36,520 crore (\$44 billion) by 2033, driven by private sector participation, gov-

ernment initiatives, and demand for satellite-based services.

Anil Prakash, Director General of SatCom Industry Association India (SIA-India), says private firms are moving the space sector into a commercialisation phase, just as SpaceX and Blue Origin did in the US.

“Innovations and investments from private sector firms are crucial to the commercialisation of space,” says Prakash. Consider the numbers: India has over 200 deep tech space start-ups, which secured \$123.90 million in funding in 2023, amassing \$380 million. In 2022, there was just one space start-up. Prakash highlights the trend towards miniaturisation and small satellites, for scientific and commercial use.

MONEY: WHENCE IT CAME, WHERE IT WENT

Invest India says the Indian satellite launch market was valued at approximately ₹11,026 crore in 2023 and is expected to expand to ₹31,804 crore by 2030, with an anticipated CAGR of around 11.7%.

Prakash attributes this growth to cost-effective launch solutions from Isro.

Skyroot Aerospace, with its Vi-

WHAT THE SECTOR LOOKS LIKE

- 1 **Indian Space Economy: Valued at ₹6,972 crore in 2023, with 2-3% of the global market share; it is expected to reach ₹36,520 crore by 2033**
- 2 **India currently has 200+ deep tech space start-ups; these secured \$123.90 million in 2023, with a total of \$380 million in funding**
- 3 **India’s satellite launch market is projected to reach ₹31,804 crore by 2030; the geospatial sector is expected to exceed ₹63,000 crore by 2025**